

Problem 3 (Bulgarian solitaire), Cell 6:
the whole function $n \mapsto D_{\mathcal{B}}(n)$

Problem 3 team

Official setting.

A partition of a positive integer n is a weakly decreasing sequence $\lambda = (\lambda_1, \dots, \lambda_s)$ of positive integers with $\lambda_1 + \dots + \lambda_s = n$. The shift $\mathcal{B}(\lambda)$ is the partition whose parts are the positive numbers among $\lambda_1 - 1, \dots, \lambda_s - 1$ together with one extra part equal to s . Call λ cyclic if $\mathcal{B}^i(\lambda) = \lambda$ for some $i \geq 1$, and let $d_{\mathcal{B}}(\lambda) = \min\{i \geq 0 : \mathcal{B}^i(\lambda) \text{ is cyclic}\}$, $D_{\mathcal{B}}(n) = \max\{d_{\mathcal{B}}(\lambda) : \lambda \text{ is a partition of } n\}$. Write $T_k = k(k+1)/2$ and $\delta_k = (k, k-1, \dots, 2, 1)$. Every $n \geq 1$ satisfies $T_{k-1} < n \leq T_k$ for exactly one k , which we call the rank of n .

Official statement of the cell (13 points, open question).

Finally, the whole function $n \mapsto D_{\mathcal{B}}(n)$. Determine $D_{\mathcal{B}}(n)$ for every n .

Status. Partial. The cell asks for a closed form for $D_{\mathcal{B}}(n)$ valid for *every* n ; we do not have one, and we do not claim one. What this document contains is, precisely:

- (a) *Proved, for all k in the stated ranges:* the four columns $r \in \{1, 2, k-1, k\}$ of the array $D_{\mathcal{B}}(T_{k-1} + r)$ (Cells 2–5), the general upper bound $D_{\mathcal{B}}(n) \leq k^2 - 2k - 1$ for non-triangular n (Cell 3), and one new step proved here: the exact value $d_{\mathcal{B}}(1^n) = n - k + 1$ for every $n \geq 3$, which gives the lower bound of Conjecture 2.1 in the two *middle* columns $r \in \{a, a+1\}$, $a = \lfloor (k-1)/2 \rfloor$ (Theorem 5.3, Corollary 5.4).
- (b) *Computer-checked only:* $D_{\mathcal{B}}(n)$ is computed exhaustively and *exactly* for every $n \leq 80$, and it agrees with the Griggs–Ho Conjecture 2.1 formula for every n with $3 \leq n \leq 80$. There is **no disagreement** anywhere in that range. This covers all ranks $k \leq 12$ (i.e. past $T_{11} = 66$ and $T_{12} = 78$). Wall clock and hardware are reported in Section 4; the run is a single exhaustive pass, 300.8 s on a laptop.
- (c) *Open:* the columns $3 \leq r \leq k-2$ other than the two middle ones, and the *upper* bound in the two middle columns. Equivalently: Conjecture 2.1 is now reduced, in the middle columns, to an upper bound, and is untouched by us elsewhere in that range.

Section 7 states this again, item by item, at the end.

Answer, as far as it goes. Write $n = T_{k-1} + r$ with $1 \leq r \leq k$ and put $a = \lfloor (k-1)/2 \rfloor$. The conjectured answer, which we confirm for all $n \leq 80$ and prove in the columns listed above, is

$$D_{\mathcal{B}}(T_{k-1} + r) = \begin{cases} (k-3-r)k + r + 2, & 1 \leq r \leq a-1, \\ n - k + 1 = T_{k-1} + r - k + 1, & r \in \{a, a+1\}, \\ (r-2)k + r, & a+2 \leq r \leq k. \end{cases}$$

1 Setting, notation, and what we quote

Throughout, $T_m = m(m+1)/2$ for $m \geq 0$ (so $T_0 = 0$), and a *partition* of $n \geq 1$ is a finite non-increasing sequence of positive integers with sum n ; we identify partitions with multisets of positive integers, which is legitimate because $\mathcal{B}$ depends only on the multiset. We write $\ell(\lambda)$ for the number of parts, $\delta_m = (m, m-1, \dots, 1)$, and $1^n = (1, 1, \dots, 1)$ (n parts). For a partition λ with $\ell(\lambda) = s$,

$$\mathcal{B}(\lambda) = \{\lambda_i - 1 : \lambda_i \geq 2\} \cup \{s\} \quad (\text{as a multiset}).$$

Since there are finitely many partitions of n and $\mathcal{B}$ maps partitions of n to partitions of n , every orbit $\lambda, \mathcal{B}(\lambda), \mathcal{B}^2(\lambda), \dots$ is eventually periodic, so $d_{\mathcal{B}}(\lambda)$ is well defined; $d_{\mathcal{B}}(\lambda)$ is exactly the length of the tail of that orbit, i.e. the distance from λ to the set of cyclic partitions in the functional graph of $\mathcal{B}$. Every $n \geq 1$ has a unique *rank* k with $T_{k-1} < n \leq T_k$, and we always write $n = T_{k-1} + r$ with $1 \leq r \leq k$. Full proofs of these two facts are Lemma 1.1 and Lemma 1.2 of `P3_C2/p3_c2.tex`, and are repeated verbatim in Cells 3–5; we do not repeat them here.

The only external input we use is Brandt’s description of the cyclic partitions.

Theorem 1.1 (Brandt [Brandt82]; see also [GH98, §2]). *Let $n = T_{k-1} + r$ with $1 \leq r \leq k$. A partition $\lambda \vdash n$ is cyclic if and only if*

$$\lambda = \{i + e_i : 0 \leq i \leq k-1, i + e_i > 0\}, \quad e_i \in \{0, 1\}, \quad \sum_{i=0}^{k-1} e_i = r.$$

In particular every cyclic partition of n has all parts $\leq k$, has k or $k-1$ parts, and $\mathcal{B}$ permutes this set cyclically in orbits whose lengths divide k .

A proof is in `P3_C2/p3_c2.tex`, Theorem 2.6 (repeated as Theorem 2.6 of Cells 3–5). We use only the displayed characterisation and the consequence “*cyclic* $\Rightarrow$ *all parts* $\leq k$ ”.

2 The conjectured answer

Griggs and Ho [GH98, Thm. 4.5] construct, for every $n \geq 3$, a partition of n with a prescribed value of $d_{\mathcal{B}}$, obtaining a lower bound for $D_{\mathcal{B}}(n)$; they conjecture that this lower bound is the truth.

Conjecture 2.1 (Griggs–Ho [GH98, Conj. 4.7]). Let $n \geq 3$, let k be the rank of n , write $n = T_{k-1} + r$ with $1 \leq r \leq k$, and put $a = \lfloor (k-1)/2 \rfloor$. Then

$$D_{\mathcal{B}}(n) = G(n) := \begin{cases} (k-3-r)k + r + 2, & 1 \leq r \leq a-1, \\ n - k + 1, & r \in \{a, a+1\}, \\ (r-2)k + r, & a+2 \leq r \leq k. \end{cases}$$

The three branches are exactly the three witness families of [GH98, Thm. 4.5]; the inequality $D_{\mathcal{B}}(n) \geq G(n)$ is their theorem, the equality is the conjecture. Griggs and Ho verified the equality for $n \leq 36$ and proved it for $r \in \{k-1, k\}$. Two remarks on the shape of G , both immediate from the display: G decreases by $k-1$ as r increases by 1 in the first branch, is flat-to-+1 across the two middle values, and increases by $k+1$ per step in the third branch; and the three branches agree with the four proved columns recalled in Section 3 (e.g. $r = k$ gives $(k-2)k + k = k^2 - k$, $r = k-1$ gives $(k-3)k + k - 1 = k^2 - 2k - 1$, $r = 1$ gives $(k-4)k + 3 = (k-1)(k-3)$ when $a \geq 2$, $r = 2$ gives $(k-5)k + 4 = (k-1)(k-4)$ when $a \geq 3$).

3 What is already proved, and by which cell

The four extreme columns of the array $D_{\mathcal{B}}(T_{k-1} + r)$ are settled in Cells 2–5, with complete proofs of both bounds. We restate them here in the (k, r) coordinates of this cell; no proof is repeated, each is a pointer.

Theorem 3.1 (Cell 2, $r = k$: triangular n). $D_{\mathcal{B}}(T_k) = k^2 - k$ for every $k \geq 1$.

(P3_C2/p3_c2.tex, Theorem 5.1; lower bound by the explicit family $\lambda^{[k]} = (k-1, k-1, k-2, \dots, 2, 1, 1)$, upper bound the Igusa–Etienne theorem as proved there following [GH98, Thm. 3.7].)

Theorem 3.2 (Cell 3, general upper bound and $r = k-1$). For every $k \geq 4$ and every non-triangular n with $T_{k-1} < n < T_k$ one has $D_{\mathcal{B}}(n) \leq k^2 - 2k - 1$; and $D_{\mathcal{B}}(T_k - 1) = k^2 - 2k - 1$ for every $k \geq 4$, attained by $\lambda^{(k)} = (k-1, k-2, k-2, k-3, \dots, 2, 1, 1)$.

(P3_C3/p3_c3.tex, Theorems 5.1 and 5.2, following [GH98, Thm. 4.4].) This is the only bound we have that applies to *all* the columns $1 \leq r \leq k-1$ at once, and it is exactly the obstacle: it is sharp only at $r = k-1$, and for r in the middle it overshoots $G(n)$ by roughly $k^2/2$.

Theorem 3.3 (Cell 4, $r = 1$). $D_{\mathcal{B}}(T_{k-1} + 1) = (k-1)(k-3)$ for every $k \geq 5$.

(P3_C4/p3_c4.tex, Corollary 6.2. This is the case $r = 1$ of Conjecture 2.1.)

Theorem 3.4 (Cell 5, $r = 2$). $D_{\mathcal{B}}(T_{k-1} + 2) = (k-1)(k-4)$ for every $k \geq 7$, and $D_{\mathcal{B}}(T_{k-1} + 2) = \max\{(k-1)(k-4), T_{k-2} + 2\}$ for every $k \geq 2$.

(P3_C5/p3_c5.tex, Theorem 6.1. This is the case $r = 2$ of Conjecture 2.1; note that for $3 \leq k \leq 6$ one has $r = 2 \in \{a, a+1\}$, and the second branch of G gives $T_{k-2} + 2 = n - k + 1$, so the two displays agree; for $k = 2$ the third branch gives $D_{\mathcal{B}}(3) = 2$, again as stated.)

Corollary 3.5. Conjecture 2.1 holds for $r \in \{1, 2, k-1, k\}$ (in the ranges of Theorems 3.1–3.4); for every k this is 4 of the k columns.

4 The exhaustive computation

What is computed. For each n in turn, the script `p3_c6_DB.py` enumerates *all* $p(n)$ partitions of n , builds the functional graph of $\mathcal{B}$ on them, finds the cyclic partitions as the vertices lying on a cycle of that graph (Brandt’s theorem is *not* used: the cyclic set is found by the definition, as a vertex λ with $\mathcal{B}^i(\lambda) = \lambda$ for some $i \geq 1$), computes $d_{\mathcal{B}}(\lambda)$ for every λ as the distance to that set, and returns $D_{\mathcal{B}}(n) = \max_{\lambda} d_{\mathcal{B}}(\lambda)$. It then compares with $G(n)$.

Why it gets past $n = 60$. The version used in Cells 2–5 (`bs.py`) stores every partition as a tuple, in a dictionary and in several per-partition lists, which is what stops it around $n = 60$. Here a partition is stored as its integer rank: with $P[m][M] = \#\{\mu \vdash m : \mu_1 \leq M\}$ precomputed, listing the partitions of m with parts $\leq M$ by decreasing first part lets one skip a whole block in $O(1)$, so a partition is ranked in $O(\ell(\lambda))$ time and the graph is two integer arrays of length $p(n)$.

Honest statement of the range and the timing. The run reported in `p3.c6.DB.log` covers $1 \leq n \leq 80$ and took 300.8s of **wall clock** on the laptop used for the hand-in (Apple M3 Max, CPython 3.12, one core), i.e. inside the 10-minute limit of the hand-in rules. The script takes a time budget as its second argument and stops before starting an n that would break it, so the range is reproducible. The last few values of n dominate: $n = 80$ alone accounts for about 40s. Going further is only a matter of budget, not of method, but $p(n)$ grows by a factor ≈ 1.15 per unit of n , so each further +5 in n roughly doubles the run.

Cross-check. For $n \leq 28$ the same run recomputes $D_{\mathcal{B}}(n)$ by the naive method (iterate $\mathcal{B}$ from every single partition until a partition repeats, straight from the definition, as in `p3.small_cases.py`) and checks that the two agree; they do. Independently, all values with $n \leq 60$ agree with the earlier `bs.py` run recorded in `bs_output_60.txt`, which was produced by a different implementation and is also cross-checked there against Brandt’s theorem, against the counts of cycles, and against the witnesses of Cells 2–5.

Theorem 4.1 (computer-checked). *For every n with $3 \leq n \leq 80$, $D_{\mathcal{B}}(n) = G(n)$. In particular Conjecture 2.1 holds for all $n \leq 80$, i.e. for all ranks $k \leq 12$, and **no disagreement with the Griggs–Ho formula was found anywhere in that range.***

Proof. Exhaustive computation; see the discussion above and the log `p3.c6.DB.log`. This is a machine verification, not a proof in the mathematical sense, and it is used nowhere in Sections 3 and 5. $\square$

The values, arranged as the array $D_{\mathcal{B}}(T_{k-1} + r)$ (row k , column $r = 1, \dots, k$):

$D_{\mathcal{B}}$	$r = 1$	$r = 2$	$r = 3$	$r = 4$	$r = 5$	$r = 6$	$r = 7$	$r = 8$	$r = 9$	$r = 10$	$r = 11$	$r = 12$
$k = 1$	0											
$k = 2$	0	2										
$k = 3$	2	3	6									
$k = 4$	4	5	7	12								
$k = 5$	8	8	9	14	20							
$k = 6$	15	12	13	16	23	30						
$k = 7$	24	18	18	19	26	34	42					
$k = 8$	35	28	24	25	29	38	47	56				
$k = 9$	48	40	32	32	33	42	52	62	72			
$k = 10$	63	54	45	40	41	46	57	68	79	90		
$k = 11$	80	70	60	50	50	51	62	74	86	98	110	
$k = 12$	99	88	77	66	60	61	67	80	93	106	119	132

Read along a row: the value falls by $k - 1$ per step, flattens at the two middle entries (boxed), then rises by $k + 1$ per step. The four columns proved in Cells 2–5 are $r = 1, 2$ (left) and $r = k - 1, k$ (right end of each row); Section 5 adds the lower bound at the two boxed entries.

5 One new step: the orbit of 1^n and the middle columns

The middle branch $G(n) = n - k + 1$ of Conjecture 2.1 has a witness that is much simpler than the witnesses of the other two branches, and it can be analysed completely. Everything in this section is proved; nothing in it uses the computation of Section 4.

Lemma 5.1 (one step of the staircase). *Let $j \geq 0$ and $m \geq j + 2$ be integers. Then, as multisets,*

$$\mathcal{B}^{j+1}(\{m\} \cup \delta_j) = \{m - j - 1\} \cup \delta_{j+1},$$

and for $1 \leq i \leq j$ the intermediate partitions are

$$\mathcal{B}^i(\{m\} \cup \delta_j) = \{m - i\} \cup \{j + 1\} \cup (\delta_j \setminus \{j + 1 - i\}). \quad (S_i)$$

(Here $\delta_0 = \emptyset$, and $\delta_j \setminus \{x\}$ means: delete one part equal to x .)

Proof. Put $S_0 = \{m\} \cup \delta_j$ and let S_i be the right-hand side of (S_i) for $1 \leq i \leq j$, and $S_{j+1} = \{m - j - 1\} \cup \delta_{j+1}$. Each S_i with $0 \leq i \leq j$ has exactly $j + 1$ parts: S_0 has $1 + j$; and for $1 \leq i \leq j$ it has $1 + 1 + (j - 1) = j + 1$ parts, because $1 \leq j + 1 - i \leq j$ so exactly one part is deleted from δ_j . All parts are positive: the only part that could vanish is $m - i \geq m - j - 1 \geq 1$. It therefore suffices to check $\mathcal{B}(S_i) = S_{i+1}$ for $0 \leq i \leq j$, and $\mathcal{B}$ on a partition with $j + 1$ parts subtracts 1 from every part, deletes the resulting zeros, and adjoins one part equal to $j + 1$.

$i = 0$: subtracting 1 turns $\{m\} \cup \delta_j$ into $\{m - 1\} \cup \{j - 1, j - 2, \dots, 1, 0\}$; deleting the 0 and adjoining $j + 1$ gives $\{m - 1\} \cup \{j + 1\} \cup (\delta_j \setminus \{j\})$, which is S_1 .

$1 \leq i \leq j - 1$: subtracting 1 turns $\{m - i\} \cup \{j + 1\} \cup (\{j, \dots, 1\} \setminus \{j + 1 - i\})$ into $\{m - i - 1\} \cup \{j\} \cup (\{j - 1, \dots, 0\} \setminus \{j - i\})$. Since $i \leq j - 1$ we have $j - i \geq 1$, so the part 0 is present exactly once and is deleted, leaving $\{m - i - 1\} \cup \{j\} \cup (\{j - 1, \dots, 1\} \setminus \{j - i\})$; adjoining $j + 1$ gives $\{m - i - 1\} \cup \{j + 1\} \cup (\{j, \dots, 1\} \setminus \{j - i\}) = S_{i+1}$, as $j - i \leq j - 1$.

$i = j$: here $S_j = \{m - j\} \cup \{j + 1\} \cup \{j, j - 1, \dots, 2\}$ (the part 1 is the deleted one). Subtracting 1 gives $\{m - j - 1\} \cup \{j\} \cup \{j - 1, \dots, 1\}$, with no zero to delete; adjoining $j + 1$ gives $\{m - j - 1\} \cup \delta_{j+1} = S_{j+1}$. $\square$

Lemma 5.2 (the staircase orbit of a single pile). *Let $n \geq 1$ and let $j \geq 0$ be such that $n > T_j$. Then $\mathcal{B}^{T_j}((n)) = \{n - T_j\} \cup \delta_j$.*

Proof. Induction on j , the case $j = 0$ being $\mathcal{B}^0((n)) = (n)$. Assume the statement for j and let $n > T_{j+1}$. Then also $n > T_j$, so $\mathcal{B}^{T_j}((n)) = \{m\} \cup \delta_j$ with $m = n - T_j > T_{j+1} - T_j = j + 1$, i.e. $m \geq j + 2$. Lemma 5.1 gives $\mathcal{B}^{T_j+j+1}((n)) = \{m - j - 1\} \cup \delta_{j+1}$, and $T_j + j + 1 = T_{j+1}$, $m - j - 1 = n - T_{j+1}$. $\square$

Theorem 5.3. *Let $n \geq 3$, let k be its rank and write $n = T_{k-1} + r$ with $1 \leq r \leq k$. Then*

$$d_{\mathcal{B}}((n)) = n - k \quad \text{and} \quad d_{\mathcal{B}}(1^n) = n - k + 1.$$

Moreover $\mathcal{B}^{n-k}((n)) = \delta_k \setminus \{k - r\}$ (which for $r = k$ reads δ_k).

Proof. Note first $k \geq 2$ and $n - k = T_{k-2} + (r - 1)$, because $T_{k-1} - (k - 1) = T_{k-2}$.

Step 1: identification of $\mathcal{B}^{n-k}((n))$. Since $n > T_{k-1} \geq T_{k-2}$, Lemma 5.2 gives $\mathcal{B}^{T_{k-2}}((n)) = \{m\} \cup \delta_{k-2}$ with $m = n - T_{k-2} = k - 1 + r$. As $m \geq k = (k - 2) + 2$, we may apply Lemma 5.1 with $j = k - 2$. If $1 \leq r - 1 \leq k - 2$, formula (S_{r-1}) with $i = r - 1$ gives

$$\mathcal{B}^{T_{k-2}+r-1}((n)) = \{k\} \cup \{k - 1\} \cup (\delta_{k-2} \setminus \{k - r\}) = \delta_k \setminus \{k - r\};$$

if $r - 1 = 0$ it is $S_0 = \{k\} \cup \delta_{k-2} = \delta_k \setminus \{k-1\}$, the same formula; and if $r - 1 = k - 1$ (i.e. $r = k$, $n = T_k$) it is $S_{(k-2)+1} = \{m - k + 1\} \cup \delta_{k-1} = \{k\} \cup \delta_{k-1} = \delta_k$, again the same formula with $k - r = 0$. In all cases $\mathcal{B}^{n-k}((n)) = \delta_k \setminus \{k - r\}$.

Step 2: this partition is cyclic. Apply Theorem 1.1 with $e_i = 1$ for $k - r \leq i \leq k - 1$ and $e_i = 0$ for $0 \leq i \leq k - r - 1$ (so $\sum e_i = r$): the resulting multiset is $\{k, k - 1, \dots, k - r + 1\} \cup \{k - r - 1, k - r - 2, \dots, 1\} = \delta_k \setminus \{k - r\}$. Hence it is cyclic.

Step 3: nothing earlier is cyclic. We show that $\mathcal{B}^t((n))$ has a part $\geq k + 1$ for every $0 \leq t < n - k$; by Theorem 1.1 such a partition cannot be cyclic. Let $t < n - k = T_{k-2} + r - 1$.

If $T_{k-2} \leq t$, write $t = T_{k-2} + i$ with $0 \leq i < r - 1$; by Step 1 the partition $\mathcal{B}^t((n))$ is S_i (in the notation of Lemma 5.1 with $j = k - 2$, $m = k - 1 + r$), which contains the part $m - i = k - 1 + r - i \geq k + 1$.

If $t < T_{k-2}$, let $j \leq k - 3$ be maximal with $T_j \leq t$ and put $i = t - T_j$, so $0 \leq i \leq j$ (indeed $t < T_{j+1} = T_j + j + 1$). Since $n > T_{k-1} > T_j$, Lemma 5.2 applies and $\mathcal{B}^{T_j}((n)) = \{m'\} \cup \delta_j$ with $m' = n - T_j \geq j + 2$; by Lemma 5.1, $\mathcal{B}^t((n))$ contains the part $m' - i = n - T_j - i = n - t$. As $t < T_{k-2}$ we get $n - t > n - T_{k-2} = k - 1 + r \geq k$, i.e. $n - t \geq k + 1$.

Hence $d_{\mathcal{B}}((n)) = n - k$.

Step 4: from (n) to 1^n . $\mathcal{B}(1^n) = (n)$, because every part 1 becomes 0 and is deleted and one part equal to $\ell(1^n) = n$ is adjoined. Since $n \geq 3$ we have $n > k$, so $d_{\mathcal{B}}((n)) = n - k \geq 1$ and (n) is not cyclic; a fortiori 1^n is not cyclic, since $\mathcal{B}$ maps cyclic partitions to cyclic partitions. Therefore $d_{\mathcal{B}}(1^n) = 1 + d_{\mathcal{B}}((n)) = n - k + 1$. $\square$

Corollary 5.4. *For every $n \geq 3$ of rank k , $D_{\mathcal{B}}(n) \geq n - k + 1$. In particular, for $r \in \{a, a + 1\}$ with $a = \lfloor (k - 1)/2 \rfloor$, the lower bound of Conjecture 2.1 holds: $D_{\mathcal{B}}(T_{k-1} + r) \geq G(T_{k-1} + r) = n - k + 1$. Consequently, in the two middle columns Conjecture 2.1 is equivalent to the single inequality $D_{\mathcal{B}}(n) \leq n - k + 1$.*

Proof. Immediate from Theorem 5.3 and the definition of $D_{\mathcal{B}}$, together with the middle branch of G . $\square$

Remark 5.5. What this does *not* do. The upper bound in the middle columns is not proved, and it is the hard half: the only general upper bound available is $k^2 - 2k - 1$ (Theorem 3.2), whereas $n - k + 1 \approx k^2/2$, so the gap is about $k^2/2$ — of the same order as the answer. The Cell 4 and Cell 5 upper bounds do not extend: their entry lemmas classify how an orbit enters the cycle using that the cyclic partitions of $T_{k-1} + r$ are indexed by r -subsets of $\{0, \dots, k - 1\}$, and for $r \approx k/2$ the number of entry types is $\binom{k}{r}$, which is exponential in k , not $O(k)$ as for $r = 1, 2$. Section 7 says where we think the next step lies.

Remark 5.6 (relation to the literature). Theorem 5.3 is the third witness family of [GH98, Thm. 4.5], for which they state the value without a detailed proof of the tail length; we give a complete proof here, and the exact identification $\mathcal{B}^{n-k}((n)) = \delta_k \setminus \{k - r\}$ of the cyclic partition that the single pile falls into appears to be slightly more precise than what is stated there. We make no novelty claim beyond that: the same caveat as in Cells 4 and 5 applies (we checked the sources cited here and three later ones, but did not survey the literature exhaustively).

6 Numerical evidence (not proofs)

Everything in this section is an observation about the computed range $n \leq 80$ only. It is recorded because it says where to look next, and it is kept strictly separate from Sections 3 and 5.

Numerical observation 6.1. For $4 \leq n \leq 80$, the partition 1^n is extremal (i.e. $d_{\mathcal{B}}(1^n) = D_{\mathcal{B}}(n)$) *exactly* when $r \in \{a, a+1\}$; the single exception $n = 3$ is an accidental coincidence of two branches of G . The extremal *sets* were computed only for $n \leq 60$ (`bs_output_60.txt`); there, 1^n is the *unique* extremal partition for every k with $5 \leq k \leq 10$ in the column $r = a+1$ ($n = 13, 18, 25, 32, 41, 50$), and also in the column $r = a$ when k is even ($n = 17, 31, 49$), whereas for $r = a$ with k odd the extremal set is large (2 partitions for $n = 12$, 31 for $n = 24$, 821 for $n = 40$, 16653 for $n = 60$).

This is the sharpest structural hint we have: at $r = a+1$ the maximum is attained only by the single “flattest” partition, so an upper-bound proof there should be able to argue by “any partition other than 1^n loses at least one step”, which is not how the Cell 2–5 upper bounds are organised.

Numerical observation 6.2. The columns $r = 3$ and $r = k-2$, the natural next targets, behave exactly as G predicts: $D_{\mathcal{B}}(T_{k-1} + 3) = (k-6)k + 5 = (k-1)(k-5)$ for all k with $9 \leq k \leq 12$ (and $D_{\mathcal{B}}(T_{k-1} + 3) = n - k + 1$ for $5 \leq k \leq 8$, where $3 \in \{a, a+1\}$), and $D_{\mathcal{B}}(T_k - 2) = (k-4)k + k - 2 = k^2 - 3k - 2$ for all k with $6 \leq k \leq 12$. We did not find a proof of either within this cell.

Numerical observation 6.3. The Cell 3 bound $k^2 - 2k - 1$ is attained only at $r = k-1$; the deficit $k^2 - 2k - 1 - G(n)$ is 0 at $r = k-1$, $k+1$ at $r = k-2$, and grows to roughly $k^2/2$ in the middle of the row. So a proof of Conjecture 2.1 cannot be a perturbation of the Cell 3 argument except very near the right end of the row; this matches what Cell 5 already found for $r = 2$ at the left end.

7 Status: partial

Proved (no computer used).

- $D_{\mathcal{B}}(T_k) = k^2 - k$, all $k \geq 1$ (Theorem 3.1, Cell 2).
- $D_{\mathcal{B}}(n) \leq k^2 - 2k - 1$ for all non-triangular n of rank $k \geq 4$, and $D_{\mathcal{B}}(T_k - 1) = k^2 - 2k - 1$ for all $k \geq 4$ (Theorem 3.2, Cell 3).
- $D_{\mathcal{B}}(T_{k-1} + 1) = (k-1)(k-3)$, all $k \geq 5$ (Theorem 3.3, Cell 4).
- $D_{\mathcal{B}}(T_{k-1} + 2) = \max\{(k-1)(k-4), T_{k-2} + 2\}$, all $k \geq 2$ (Theorem 3.4, Cell 5).
- New here: $d_{\mathcal{B}}((n)) = n - k$ and $d_{\mathcal{B}}(1^n) = n - k + 1$ for every $n \geq 3$, with the explicit landing point $\mathcal{B}^{n-k}((n)) = \delta_k \setminus \{k-r\}$ (Theorem 5.3); hence $D_{\mathcal{B}}(n) \geq n - k + 1$ always, which is the lower half of Conjecture 2.1 in the two middle columns $r \in \{a, a+1\}$ (Corollary 5.4).

Small values used by Cells 3 and 5 for a bounded set of k (namely $k = 4$ in Cell 3 and $2 \leq k \leq 6$ in Cell 5) are exhaustive computations, as declared in those cells.

Computer-checked only. $D_{\mathcal{B}}(n)$ is known exactly for every $n \leq 80$, and equals $G(n)$ for every n with $3 \leq n \leq 80$ (Theorem 4.1); no disagreement with Conjecture 2.1 occurs in that range. This is an exhaustive computation over all $p(n)$ partitions, 300.8s of wall clock in total, cross-checked against a naive implementation for $n \leq 28$ and against the independent `bs.py` run for $n \leq 60$. Nothing in Sections 3 and 5 depends on it.

Open.

- The value of $D_{\mathcal{B}}(T_{k-1} + r)$ for $3 \leq r \leq k - 2$ with $r \notin \{a, a + 1\}$: nothing is proved, for any single such (k, r) with k unbounded.
- The upper bound $D_{\mathcal{B}}(T_{k-1} + r) \leq n - k + 1$ for $r \in \{a, a + 1\}$: the lower bound is now proved (Corollary 5.4), the upper bound is not.
- Therefore Conjecture 2.1, hence the official question of this cell, is **not** answered. We have 4 columns out of k proved, 2 more with the lower bound proved, and the whole array confirmed numerically up to $n = 80$.

Where we would go next. Observation 6.1 (uniqueness of the extremal partition at $r = a + 1$) suggests attacking $r = a + 1$ before $r = 3$: uniqueness usually means a rigid orbit, and the orbit of 1^n is completely described by Lemma 5.1. The obstruction identified in the Remark after Corollary 5.4 — that the entry classification used in Cells 4 and 5 has $\binom{k}{r}$ types — is what has to be replaced, presumably by an argument that bounds the tail by a potential function on the orbit rather than by its entry point.

A Code

`p3_c6_DB.py` (delivered with this document, together with its output `p3_c6_DB.log`) computes $D_{\mathcal{B}}(n)$ exhaustively for $n = 1, \dots, 80$ as described in Section 4, compares each value with $G(n)$, cross-checks against a naive orbit-by-orbit computation for $n \leq 28$, and prints the array of Section 4. Usage: `python3 p3_c6_DB.py [NMAX] [BUDGET_SECONDS]`; the run delivered here is `python3 p3_c6_DB.py 80 480`, wall clock 300.8s. The earlier and independent `bs.py` (delivered with Cells 2–5, output `bs_output.60.txt`) covers $n \leq 60$ and additionally verifies Brandt’s theorem, the cycle counts, and every witness and characterisation used in Cells 2–5.

References

- [Brandt82] J. Brandt, Cycles of partitions, *Proc. Amer. Math. Soc.* **85** (1982), 483–486.
- [GH98] J. R. Griggs and C.-C. Ho, The cycling of partitions and compositions under repeated shifts, *Adv. Appl. Math.* **21** (1998), 205–227.